

Ekaterine Dadiani

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EDUCATION

Ph.D. Candidate in Physics – Carnegie Mellon University GPA for Spring 2025 – 3.56 out of 4.0 Research topic: Studying LISA sources using DESI and LIGO/Virgo/KAGRA Advisors: Antonella Palmese, Tiziana Di Matteo	2023 – Present Pittsburgh, PA
M.Sc. in Physics – Carnegie Mellon University GPA – 3.51 out of 4.0 Research topic: Morphology of $z \sim 2$ Dual AGN Host Galaxies in ASTRID Simulation Advisor: Tiziana Di Matteo	2021 – 2023 Pittsburgh, PA
B.Sc. in Physics – Ivane Javakhishvili Tbilisi State University GPA – 4.0 out of 4.0 Bachelor's thesis: "Transient Flow and Magnetic Field Generation in 2 Temperature Degenerate Electron-Ion Gas due to Catastrophic Relaxation of Equilibrium" Advisor: Nana Shatashvili	2016 – 2020 Tbilisi, Georgia
199th Physical-Mathematical School Named after V. Komarovi GPA 4.0/4.0, Graduated with honors (Gold Medal)	2010 – 2016 Tbilisi, Georgia
Technological Center "MZIURI" A two-year program focused on learning the basics of C++ and logic	2009 – 2010 Tbilisi, Georgia

GRANTS & FELLOWSHIPS

NASA LISA Preparatory Science Grant PI: Antonella Palmese, Tiziana Di Matteo "Studying LISA sources using DESI and LIGO/Virgo/KAGRA", \$687k.	2023–2026
LSST Discovery Alliance Data Science Fellowship A two-year fellowship offering intensive training in Software Engineering, Statistics, Machine Learning, Scalable Programming and Data Management, Time Series Analysis, Science Communication, etc.	2025–2026
Data Science Pathway of the CI Pathways Program <i>NSF Award 2417789; Led by NCSA & PSC</i> A focused 10-week training program in data science, machine learning, and AI, leveraging advanced cyberinfrastructure to empower researchers with world-class computational skills.	2025

COLLABORATIONS & MEMBERSHIPS

Since 2023, I have actively contributed to two major international collaborations:

- **Dark Energy Spectroscopic Instrument (DESI)** (~ 500 members): Currently working on Project 249, "Searching for Massive Black Hole Binaries in DESI."
- **ESA/NASA Laser Interferometer Space Antenna (LISA) Consortium** ($\sim 1,000$ members): Member of the LISA Consortium within the Astrophysics Working Group at Carnegie Mellon University, focusing on modeling dual AGN host galaxies and predicting massive black hole binary events with electromagnetic counterparts.

WORK EXPERIENCE

RESEARCH ASSISTANT
Carnegie Mellon University

Jan 2024 – Present
Pittsburgh, PA

Under the supervision of **Antonella Palmese** on “identifying massive black hole binary candidates using DESI spectra.” Analyzing Doppler-induced velocity shifts in broad spectral lines to detect orbital motion and applying machine learning models (including autoencoders) to extract subtle patterns from large datasets. These binaries are potential LISA-detectable gravitational wave sources that play a critical role in our understanding of the evolution of the universe.

Under the supervision of **Tiziana Di Matteo** on “the morphological analysis of dual AGN host galaxies in cosmological simulations”. Investigated the relation between galaxy morphology and stellar kinematics, identifying disk structures in high-redshift dual AGN host galaxies, and utilized ASTRID simulations to explore the influence of galaxy dynamics on black hole fueling and feedback processes.

TEACHING ASSISTANT
Carnegie Mellon University

Aug 2021 – Dec 2023
Pittsburgh, PA

Supported courses in Electronics 1, Physics I for Engineering Students, Physics II for Engineering Students, Matter and Interaction 1, and Matter and Interaction 2.

ASSISTANT RESEARCHER

Oct 2018 – Oct 2019
Tbilisi, Georgia

Centre for Theoretical Astrophysics, Ilia State University

Developed self-similar MHD flow solutions to enhance coronal eruption models. Identified new classes of solutions and collaborated on refining geometric and kinematic assumptions to improve model accuracy.

VISITING SCHOLAR

Aug 2019 – Sep 2019
Leuven, Belgium

KU Leuven, Center for Mathematical Plasma Astrophysics

Conducted research on “Self-Similar Solutions for Astrophysical MHD Transients” under the supervision of Andria Rogava and Stefaan Poedts.

ASSOCIATED MEMBER OF THE PERSONNEL
CERN Summer Students Programme

Jul 2019 – Aug 2019
Geneva, Switzerland

Contributed to the BE-ABP-HSC Department under the supervision of Nicolas Mounet by developing an analytical method for computing wake functions for a circular beam pipe.

LOCAL JURY AND MEMBER OF LOC

2019

International Young Physicist Tournament (IYPT)

Tbilisi, Georgia

FOUNDER AND CEO

2015 – 2017

TeddyBot and Quark

Tbilisi, Georgia

Co-founded TeddyBot—an innovative electrical childcare device—which received recognition at Seed Forum Tbilisi and was showcased at Seed Forum Tallinn. More information: [Seed Forum Article](#)

Also led Project Quark, a guide ball based on ‘Sphero’ hardware that interacts with waypoints and retains route memory. This project secured funding offers from Georgian government organizations, including the Scientific Library of Georgia. More information: [Project Quark Funding Info](#)

PUBLICATIONS

Dadiani E., et al., “Connecting current and future dual AGN searches to LISA and PTA gravitational wave detections” (2026) – Under Collaboration Review

Chen N., Zhou Y., **Dadiani E.**, et al., “Cosmic Pairs: A DESI Census of Dual and Offset AGN as Precursors to Massive Black Hole Binaries” (2025) – <https://doi.org/10.48550/arXiv.2512.16844>

Dadiani E., et al., “ $z \sim 2$ Dual AGN Host Galaxies are Disky: Stellar Kinematics in ASTRID Simulation,” (2024) – <https://doi.org/10.33232/001c.124451>

Rogava A., Poedts S., **Dadiani E.**, “Self-Similarity for Astrophysical MHD Transients Revisited,” (2022) – <https://doi.org/10.1016/j.asr.2021.09.020>

Mounet N., Zannini C., **Dadiani E.**, et al., “Closed Form Formulas of the Indirect Space Charge Wake

Function for Axisymmetric Structures,” JACoW HB2021 (2022) – <https://cds.cern.ch/record/2841806/files/document.pdf?version=1>

Saralidze E., Shatashvili N.L., Mahajan S.M., **Dadiani E.** “*Catastrophic formation of macro-scale flow and magnetic fields in the relativistic gas of binary systems*” – <http://dx.doi.org/10.1007/s10509-025-04456-1>

CONFERENCES

- Presented a talk at *Cosmology on the Beach*, Playa del Carmen, Mexico (2024)
- Organized the *International Conference of Physics Students (ICPS)*, Tbilisi, Georgia (2024)
- Presented a talk at the *DESI Collaboration Meeting*, Marseille (2024)
- Presented a talk at the *First Pre-LISA Electromagnetic Mini-Workshop* (2023)
- Attended the *NanoGrav Workshop*, Pittsburgh (2023)
- Presented a talk at *The 5th Neighborhood Workshop*, Penn State (2023)
- Attended the *Second Michigan Cosmology Summer School* (2023)
- Presented a talk at the *Tbilisi State University VIII Faculty of Exact and Natural Science Students Conference* (2020)
- Presented a talk at the *International Conference of Physics Students (ICPS)*, Helsinki, Finland (2018)
- Presented a talk at the *Young Physicists Conference*, Tbilisi, Georgia (top prize, 2017)
- Presented a talk at the *8th Young Researchers Meeting*, Cagliari, Italy (2017)
- Presented a talk at the *7th Young Researchers Meeting*, Turin, Italy (2016)

OTHER ACHIEVEMENTS

Olympiads:

Finalist at the Intel International Science and Engineering Fair (ISEF), Arizona, Phoenix (2016).

1st Place in the “Leonardo Da Vinci” Competition (2016).

3rd Place in the “Millennium Innovative Challenge” (2016).

1st Place at the Georgian Young Physicists’ Tournament (GYPT) (2015).

1st Place at the International Young Naturalist Tournament (IYNT) (2015).

2nd Place in the Girls in Science Innovation Competition (2015).

Ranked among the top 10 at the National Educational Olympiad in Physics and Math (2011–2015).

Hackathons & Startup Competitions:

2nd Place at Garage48 (2015).

3rd Place at the Ilia State University Hardware Hackathon (2015).

3rd Place at the TSU Innovation Challenge (2016).

1st Place at FabJam Georgia (2016).

Winner of Seed Forum Georgia (2016); subsequently selected for an international forum and recognized among the best 6 companies at Seed Forum Estonia.

RESEARCH INTERESTS

Cosmological simulations; Massive black hole physics and merger dynamics; Electromagnetic counterparts of gravitational wave events; Active Galactic Nuclei and dual AGN phenomena; Galaxy morphology and evolution; Gravitational wave astrophysics; Dark Energy Spectroscopic Instrument; Synthetic observation pipelines and machine learning applications in astrophysics; ...

SKILLS / HOBBIES

Technical Skills: Python (data analysis, scientific computing, machine learning libraries such as NumPy, Pandas, scikit-learn, TensorFlow, and PyTorch), SQL, simulation and modeling (including cosmological and N-body simulations), numerical methods, and statistical analysis.

Hobbies: Film photography, skiing, hiking, ...